

Activity 6.7.3: IPv4 Address Subnetting Part 1

Learning Objectives

Upon completion of this activity, you will be able to determine network information for a given IP address and network mask.

Background

This activity is designed to teach how to compute network IP address information from a given IP address.

Scenario

When given an IP address and network mask, you will be able to determine other information about the IP address such as:

- Network address
- Network broadcast address
- Total number of host bits
- Number of hosts

Task 1: For a given IP address, Determine Network Information.

Given:

Host IP Address	172.25.114.250
Network Mask	255.255.0.0 (/16)

Find:

Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Step 1: Translate Host IP address and network mask into binary notation.

Convert the host IP address and network mask to binary:

	172	25	114	250
IP Address	10101100	11001000	01110010	11111010
Network Mask	11111111	11111111	00000000	00000000
	255	255	0	0

Step 2: Determine the network address.

1. Draw a line under the mask.
2. Perform a bit-wise AND operation on the IP address and the subnet mask.

Note: 1 AND 1 results in a 1; 0 AND anything results in a 0.

3. Express the result in dotted decimal notation.
4. The result is the network address for this host IP address, which is **172.25.0.0**.

	172	25	114	250
IP Address	10101100	11001000	01110010	11111010
Subnet Mask	11111111	11111111	00000000	00000000
Network Address	10101100	11001000	00000000	00000000
	172	25	0	0

Step 3: Determine the broadcast address for the network address

The network mask separates the network portion of the address from the host portion. The network address has all 0s in the host portion of the address and the broadcast address has all 1s in the host portion of the address.

	172	25	0	0
Network Add.	10101100	11001000	00000000	00000000
Mask	11111111	11111111	00000000	00000000
Broadcast.	10101100	11001000	11111111	11111111
	172	25	255	255

By counting the number of host bits, we can determine the total number of usable hosts for this network.

Host bits: 16

Total number of hosts:

$$2^{16} = 65,536$$

65,536 – 2 = 65,534 (addresses that cannot use the *all 0s* address, network address, or the *all 1s* address, broadcast address.)

Add this information to the table:

Host IP Address	172.25.114.250
Network Mask	255.255.0.0 (/16)
Network Address	
Network Broadcast Address	
Total Number of Host Bits Number of Hosts	

Task 2: Challenge

For all problems:

Create a Subnetting Worksheet to show and record all work for each problem.

Problem 1

Host IP Address	172.30.1.33
Network Mask	255.255.0.0
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Problem 2

Host IP Address	172.30.1.33
Network Mask	255.255.255.0
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Problem 3

Host IP Address	192.168.10.234
Network Mask	255.255.255.0
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Problem 4

Host IP Address	172.17.99.71
Network Mask	255.255.0.0
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Problem 5

Host IP Address	192.168.3.219
Network Mask	255.255.0.0
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Problem 6

Host IP Address	192.168.3.219
Network Mask	255.255.255.224
Network Address	
Network Broadcast Address	
Total Number of Host Bits	
Number of Hosts	

Task 3: Clean Up

Remove anything that was brought into the lab, and leave the room ready for the next class.

Activity 6.7.4: IPv4 Address Subnetting Part 2

Learning Objectives

Upon completion of this activity, you will be able to determine subnet information for a given IP address and subnetwork mask.

Background

Borrowing Bits

How many bits must be borrowed to create a certain number of subnets or a certain number of hosts per subnet?

Using this chart, it is easy to determine the number of bits that must be borrowed.

Things to remember:

- Subtract 2 for the usable number of hosts per subnet, one for the subnet address and one for the broadcast address of the subnet.

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	1
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

Possible Subnet Mask Values

Because subnet masks must be contiguous 1's followed by contiguous 0's, the converted dotted decimal notation can contain one of a certain number of values:

Dec.	Binary
255	11111111
254	11111110
252	11111100
248	11111000
240	11110000
224	11100000
192	11000000
128	10000000
0	00000000

Scenario

When given an IP address, network mask, and subnetwork mask, you will be able to determine other information about the IP address such as:

- The subnet address of this subnet
- The broadcast address of this subnet
- The range of host addresses for this subnet
- The maximum number of subnets for this subnet mask
- The number of hosts for each subnet
- The number of subnet bits
- The number of this subnet

Task 1: For a Given IP Address and Subnet Mask, Determine Subnet Information.

Given:

Host IP Address	172.25.114.250
Network Mask	255.255.0.0 (/16)
Subnet Mask	255.255.255.192 (/26)

Find:

Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Step 1: Translate host IP address and subnet mask into binary notation.

	172	25	114	250
IP Address	10101100	00011001	01110010	11111010
	11111111	11111111	11111111	11000000
Subnet Mask	255	255	255	192

Step 2: Determine the network (or subnet) where this host address belongs.

1. Draw a line under the mask.
2. Perform a bit-wise AND operation on the IP Address and the Subnet Mask.
Note: 1 AND 1 results in a 1' 0 AND anything results in a 0.
3. Express the result in dotted decimal notation.

4. The result is the Subnet Address of this Subnet, which is **172.25.114.192**

	172	25	114	250
IP Address	10101100	00011001	01110010	11111010
Subnet Mask	11111111	11111111	11111111	11000000
Subnet Address	10101100	00011001	01110010	11000000
	172	25	114	192

Add this information to the table:

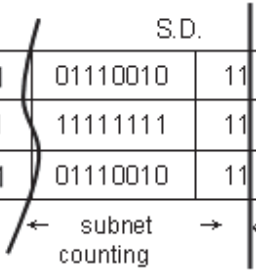
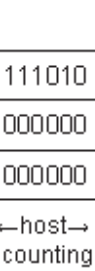
Subnet Address for this IP Address	172.25.114.192
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Step 3: Determine which bits in the address contain network information and which contain host information.

1. Draw the *Major Divide* (M.D.) as a wavy line where the 1s in the major network mask end (also the mask if there was no subnetting). In our example, the major network mask is 255.255.0.0, or the first 16 left-most bits.
2. Draw the *Subnet Divide* (S.D.) as a straight line where the 1s in the given subnet mask end. The network information ends where the 1s in the mask end.
3. The result is the Number of Subnet Bits, which can be determined by simply counting the number of bits between the M.D. and S.D., which in this case is 10 bits.

Step 4: Determine the bit ranges for subnets and hosts.

1. Label the *subnet counting range* between the M.D. and the S.D. This range contains the bits that are being incremented to create the subnet numbers or addresses.
2. Label the *host counting range* between the S.D. and the last bits at the end on the right. This range contains the bits that are being incremented to create the host numbers or addresses.

		M.D.	S.D.	
IP Address	10101110	00011001	01110010	11111010
Subnet Mask	11111111	11111111	11111111	11000000
Subnet Add.	10001010	00011001	01110010	11000000
 ← subnet counting range →  ← host counting range →				

Step 5: Determine the range of host addresses available on this subnet and the broadcast address on this subnet.

1. Copy down all of the network/subnet bits of the network address (that is, all bits before the S.D.).
2. In the host portion (to the right of the S.D.), make the host bits all 0s except for the right-most bit (or least significant bit), which you make a 1. This gives us the *first* host IP address on this

subnet, which is the *first part* of the result for *Range of Host Addresses for This Subnet*, which in the example is **172.25.114.193**.

3. Next, in the host portion (to the right of the S.D.), make the host bits all 1s except for the right-most bit (or least significant bit), which you make a 0. This gives us the *last* host IP address on this subnet, which is the last part of the result for *Range of Host Addresses for This Subnet*, which in the example is **172.25.114.254**.
4. In the host portion (to the right of the S.D.), make the host bits all 1s. This gives us the broadcast IP address on this subnet. This is the result for *Broadcast Address of This Subnet*, which in the example is **172.25.114.255**.

		M.D.	S.D.		
IP Address	10101100	00011001	01110010	11	111010
Subnet Mask	11111111	11111111	11111111	11	000000
Subnet Add.	10101100	00011001	01110010	11	000000
		~ subnet ~ counting range		~ host ~ counting range	
First Host	10101100	00011001	01110010	11	000001
	172	25	114		193
Last Host	10101100	00011001	01110010	11	111110
	172	25	114		254
Broadcast	10101100	00011001	01110010	11	111111
	172	25	114		255

Let's add some of this information to our table:

Host IP Address	172.25.114.250
Major Network Mask	255.255.0.0 (/16)
Major (Base) Network Address	172.25.0.0
Major Network Broadcast Address	172.25.255.255
Total Number of Host Bits Number of Hosts	16 bits or 2^{16} or 65,536 total hosts $65,536 - 2 = 65,534$ usable hosts
Subnet Mask	255.255.255.192 (/26)
Number of Subnet Bits Number of Subnets	
Number of Host Bits per Subnet Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Step 6: Determine the number of subnets.

The number of subnets is determined by how many bits are in the *subnet counting range* (in this example, 10 bits).

Use the formula 2^n , where n is the number of bits in the *subnet counting range*.

1. $2^{10} = 1024$

Number of Subnet Bits Number of Subnets (all 0s used, all 1s not used)	10 bits $2^{10} = 1024$ subnets
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Step 7: Determine the number usable hosts per subnet.

The number of hosts per subnet is determined by the number of host bits (in this example, 6 bits) minus 2 (1 for the subnet address and 1 for the broadcast address of the subnet).

$2^6 - 2 = 64 - 2 = 62$ hosts per subnet

Number of Host Bits per Subnet Number of Usable Hosts per Subnet	6 bits $2^6 - 2 = 64 - 2 = 62$ hosts per subnet
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Step 8: Final Answers

Host IP Address	172.25.114.250
Subnet Mask	255.255.255.192 (/26)
Number of Subnet Bits Number of Subnets	10 bits $2^{10} = 1024$ subnets
Number of Host Bits per Subnet Number of Usable Hosts per Subnet	6 bits $2^6 - 2 = 64 - 2 = 62$ hosts per subnet
Subnet Address for this IP Address	172.25.114.192
IP Address of First Host on this Subnet	172.25.114.193
IP Address of Last Host on this Subnet	172.25.114.254
Broadcast Address for this Subnet	172.25.114.255

Task 2: Challenge.

For all problems:

Create a Subnetting Worksheet to show and record all work for each problem.

Problem 1

Host IP Address	172.30.1.33
Subnet Mask	255.255.255.0
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Problem 2

Host IP Address	172.30.1.33
Subnet Mask	255.255.255.252
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Problem 3

Host IP Address	192.192.10.234
Subnet Mask	255.255.255.0
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Problem 4

Host IP Address	172.17.99.71
Subnet Mask	255.255.0.0
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Problem 5

Host IP Address	192.168.3.219
Subnet Mask	255.255.255.0
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Problem 6

Host IP Address	192.168.3.219
Subnet Mask	255.255.255.252
Number of Subnet Bits	
Number of Subnets	
Number of Host Bits per Subnet	
Number of Usable Hosts per Subnet	
Subnet Address for this IP Address	
IP Address of First Host on this Subnet	
IP Address of Last Host on this Subnet	
Broadcast Address for this Subnet	

Task 3: Clean Up

Remove anything that was brought into the lab, and leave the room ready for the next class.